Module code: BIO311
Module title: Modeling for Computational Biology
Semester 2, 2024/25, Year (4)

Final Examination

Time allowed: 3 hours

Module Examiner:
Department of Biosciences and Bioinformatics, Tel.

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INSTRUCTIONS to CANDIDATES

1. This is a **closed-book** examination, **which is to be written without books or notes.**
2. The total number of marks is 100. The numbers in brackets indicate the marks assigned to a section, question or sub-question.
2. The exam consists of four questions with various sub-questions. You need to answer **ALL** questions and all parts. There is no penalty for providing a wrong answer.
3. You need to motivate your answer when explicitly required.
4. Only English answers will be accepted. Answers should be written in the *Answer Booklet* provided.
5. Answers should be written using pen, black or blue, **NO PENCIL**.
6. All examination materials must be returned to the invigilator upon completion of the exam. Failure to do so will be deemed as academic misconduct and will be dealt with according to University's policy.

Question 1 [15 marks]

A	B	C	D	E	F	G	H	I			J
ATOM	1	N	ARG	A	20	24.557	-38.438	-2.219	1.00	48.18	N
ATOM	2	CA	ARG	A	20	24.786	-38.383	-3.724	1.00	48.20	C
ATOM	3	C	ARG	A	20	26.209	-38.780	-4.233	1.00	46.86	C
ATOM	4	O	ARG	A	20	26.312	-39.492	-5.209	1.00	46.96	O
ATOM	5	CB	ARG	A	20	24.328	-37.035	-4.295	1.00	48.44	C
ATOM	6	CG	ARG	A	20	23.791	-37.076	-5.718	1.00	49.20	C
ATOM	7	CD	ARG	A	20	23.543	-35.655	-6.309	1.00	49.60	C
ATOM	8	NE	ARG	A	20	22.419	-34.974	-5.669	1.00	54.70	N
ATOM	9	CZ	ARG	A	20	22.453	-33.754	-5.107	1.00	55.47	C
ATOM	10	NH1	ARG	A	20	23.558	-33.001	-5.109	1.00	52.88	N
ATOM	11	NH2	ARG	A	20	21.345	-33.268	-4.550	1.00	55.28	N
ATOM	12	N	SER	A	21	27.281	-38.306	-3.583	1.00	45.68	N
ATOM	13	CA	SER	A	21	28.663	-38.666	-3.946	1.00	43.64	C
ATOM	14	C	SER	A	21	29.113	-39.741	-2.991	1.00	42.98	C
ATOM	15	O	SER	A	21	28.807	-39.631	-1.810	1.00	43.65	O

Figure 1

Figure 1 shows part of a PDB file.

a) Explain what each of the column marked with a letter (A-J) represent.

[10 marks]

b) Explain why different carbon atoms in column C are labeled differently.

[5 marks]

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Question 2 [25 marks]

a) Explain what RMSD and RMSF are in the context of molecular dynamics simulations.

$$RMSD(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{r}_i(t) - \boldsymbol{r}_i(0))^2}$$

$$RMSF(i) = \sqrt{\frac{1}{T}\sum_{t=0}^{T}\left(\boldsymbol{r}_i(t) - \boldsymbol{r}_{i,AVE}\right)^2}$$

[5 marks]

b) The following graph (Figure 2) represents the RMSD profile of a protein along an MD simulation.

(b1) Provide a complete description of the graph.

(b2) Explain if we can consider the simulation well equilibrated on the basis of this graph.

[10 marks]

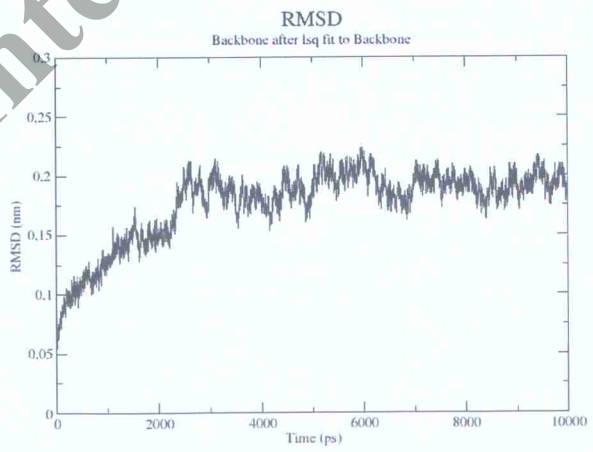


Figure 2

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c) The following graph (Figure 3) represents the RMSF profile of a protein along an MD simulation.

(c1) Provide a complete description of the graph.

(c2) Identify regions that could be alpha helixes or beta sheets, based on it.

[10 marks]

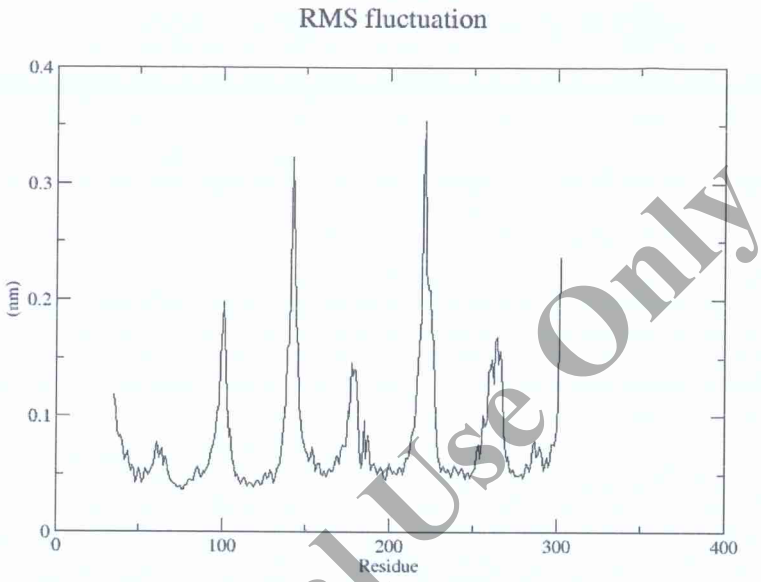


Figure 3

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Question 3 [30 marks]

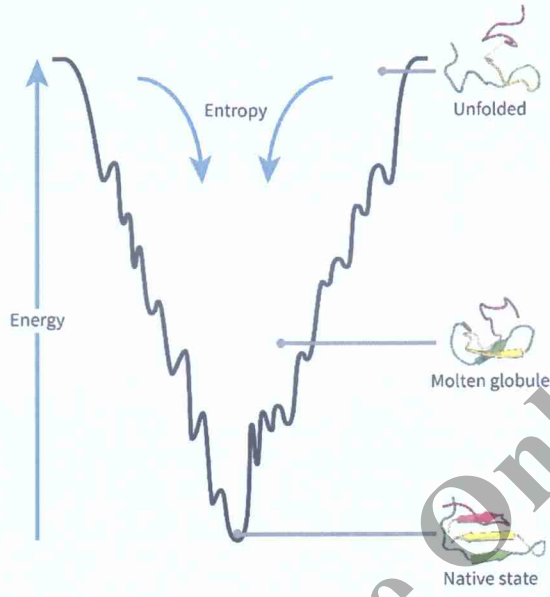


Figure 4

The Figure 4 above represents the "folding funnel" for a protein undergoing a folding process.

a) What is the relationship between the entropy of the protein and configurations it can assume? How does this entropy changes during the process?

[10 marks]

b) What contributes to the internal energy of the protein (in term of molecular interactions), and how does the internal energy change during the process?

[15 marks]

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Question 4 [30 marks]

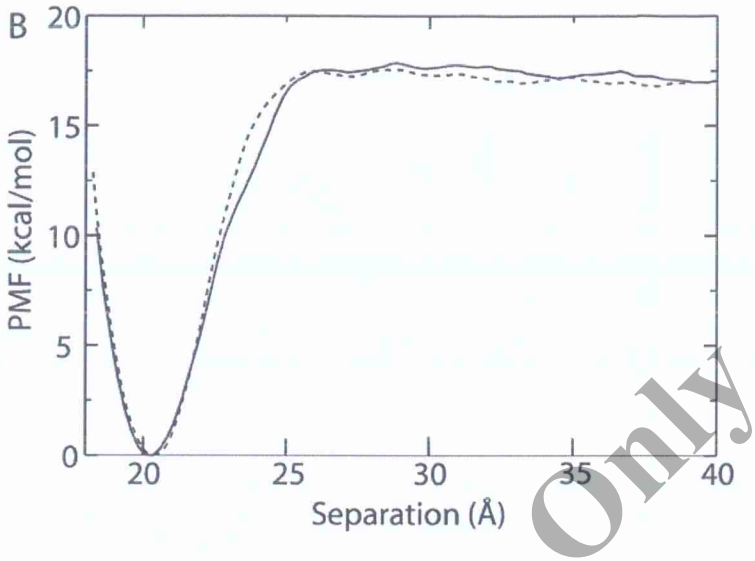


Figure 5

Figure 5 represents the Potential of Mean Force (PMF) required to separate ligand and receptor in a Steered Molecular Dynamics simulation.

a) Explain what the PMF is and how it is related to the Free Energy profile.

(You may refer to the Jarzynski equation if relevant: $\langle e^{-\frac{W}{K_B T}} \rangle = e^{-\frac{\Delta G}{K_B T}}$. However, a qualitative explanation of PMF and free energy is sufficient).

[15 marks]

b) Describe a method to calculate the PMF using MD simulations.

[15 marks]

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